



Structure, design, and test of 13 kA HTS current lead

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ABSTRACT

The 13 kA High Temperature Superconducting Current Leads (HTSCL) were designed and manufactured by the Institute of plasma physics, Chinese Academy of Sciences (ASIPP). This lead differs from former HTS current leads, in that it features an integrated heat exchanger (HEX) structure, a HTS magnetic shield and protective sleeve, and a Low Temperature Superconducting (LTS) module with a through cold conduction channel. The HTS and LTS module is cooled by SH_e. In November 2020, the 13 kA steady-state test, Loss Of Flow Accident (LOFA) test, joint resistance test, pulse current test, insulation voltage test and other low-temperature electrical test items of the leads were completed. The test results indicate that all parameters meet the design requirements.

1. Introduction

High-temperature superconducting current leads provide the electrical and thermal connection from the room-temperature termination of the power supply to the low-temperature superconducting busbar at the 4.5 K level, and they can reduce heat loads at 4.5 K and cooling power consumption about 2/3 compared with conventional current leads [1]. At the European Organization for Nuclear Research (CERN), 600 A to 12,000 A current leads have been successfully used in the Large Hadron Collider [2]. In 2002–2004, the EU Fusion Development Programme, FZK, and CRPP successfully developed and tested a first 70-kA HTS current lead as a demonstrator for ITER [3]. In 2005, a pair of 10-kA HTS current leads was developed for the 45-T hybrid magnet system in the U.S. National High Magnetic Field Laboratory [4].

In China, ASIPP has decades of experience in current lead design. The current lead design for ITER started in 2007 based on the successful delivery of 15-kA HTS current leads for the EAST tokamak [5], ITER has a total of 60 current leads, including 24 for the central solenoid (CS)/poloidal field (PF) coils (52 kA), 18 for the toroidal field (TF) coils (68 kA), and 18 for the correction coils (CC) (10 kA) [6–9]. In 2016, the three types of current lead prototypes were manufactured and tested [10–12]. In 2017, ASIPP started the series of ITER current leads production [13].

Compared with ITER HTS current lead, many new concepts and designs are added to this pair of 13 kA leads. A HEX with a fin structure integrated with the room temperature terminal was designed [14,15].

Thirty-six stacks from Sumitomo are soldered to the shunt of the HTS module. The low-temperature superconducting wire is evenly distributed on both sides of the copper block with the diversion channel to form a LTS module by soldering, and the lead is completed for low temperature electrical testing. This paper reports the structure design and test results of the 13 kA HTS trial current lead done by ASIPP feeder current lead group. The integrated HEX structure and LTS structure had been successfully applied in the East PF current leads and operate well.

2. 13 kA HTS current leads design

13KA HTS current lead is used for HTS insertion coil of large superconducting magnet in CRAFT (Comprehensive Research Facility for Fusion Technology Program of China). In the interposed coil system, the cooling medium is SH_e (see Fig. 1).

The HTS current lead is composed of a room temperature terminal, an insulating flange, a HEX, a HTS module, and a LTS module (see Fig. 2). Table 1 gives an excerpt of the specifications, listing the main design parameters.

2.1. Room temperature terminal

The structure of integrated HEX mainly includes two components: room temperature terminal and HEX (see Fig. 3). The room temperature terminal is welded with a U-shaped connecting block, and the internal water cooling channel is used to prevent freezing in the low-current or

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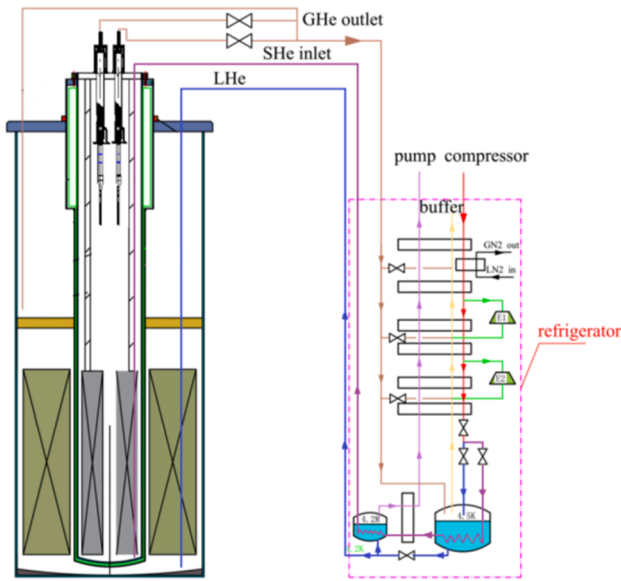


Fig. 1. Interpolation coil test platform.

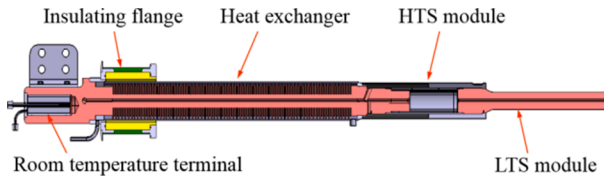


Fig. 2. Structure diagram of 13KA HTS current lead.

Table 1

Functional specifications for 13 kA leads.

Parameter	Value
Current (kA)	13
Operating temperature of top of HTS (K)	<60
LOFA time (s)	>120
Maximum design voltage (kV)	>1.5
Max joint resistance HEX-LTS (nΩ)	10
Max joint resistance LTS-busbar (nΩ)	5
Mass flow rate of HEX and 275 K HEX outlet under nominal current (g/s)	0.6 ~ 0.8
HEX pressure (Bara)	<7

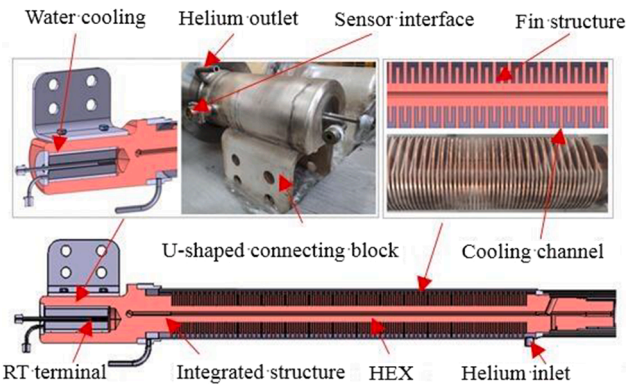


Fig. 3. Integrated heat exchanger structure of 13 kA current lead.

off-duty operation. The HEX consists of fin type structure and stainless steel main sleeve. Its performance can be controlled in a straight forward way through the manufacturing tolerances. The HEX's larger thermal

mass also slows down the temperature increase during the LOFA accident. The two ends of the stainless steel main sleeve are welded with the room temperature terminal and the outer cylinder of the HTS module respectively.

2.2. Insulating flange

During coil fast discharges, high voltage will appear on the current leads. For safety, all current leads should withstand an about 2 kV test voltage. Fig. 4 shows the insulation structure. The insulating flange is composed of an insulating sleeve, an end flange and a vacuum flange. Both ends of the insulating sleeve are sealed and connected with the end flange and the vacuum flange respectively. The sealing surfaces between the insulating sleeve and the end flange and the vacuum flange are tightly sealed by silicone rubber rings, and reinforced seal by threaded structure. A circle of insulating layer is solidified outside the insulating sleeve and between the end flange and the vacuum flange. The end flange is welded on the main bushing of the current lead, and the umbrella skirt structure is designed to effectively block the condensed water from the room temperature terminal. The design of insulation flange integrates high voltage insulation, mechanical support and vacuum sealing to meet the special requirements of electric transmission in low temperature and vacuum environment.

2.3. HTS module

The HTS module is composed of a cylindrical S.S. shunt, the hot end of HTS and 5 K end. The S.S. shunt with a total of 36 HTS stacks distributed into 18 grooves milled on its outer surface. The hot end of HTS and 5 K end are integrated with HEX and LTS respectively, and the materials are oxygen-free high-conduction copper. Each stack is soldered from five HTS tapes. The I_c of the stack is specified to be at least 600 A at 77 K in self-field (s.f.) condition. The HTS tape is from Bi-2223 AgAu tapes supplied from Sumitomo Inc., whose critical current I_c (77 K, in self field) is higher than 170 A. The main design parameters are listed in Table 2. The magnetic shielding design is added outside the shunt (see Fig. 5), which can effectively shield the influence of vertical field, and the shield sleeve is made of pure iron DT4.

2.4. LTS module

The LTS module is mainly composed of LTS copper block, LTS wire and stainless steel plate (see Fig. 6). The two ends of the LTS copper block are spherical section and square section respectively. The circular surface of the spherical section and both sides of the square section are provided with grooves for fixing the LTS wire (see Table 3). One side of the LTS wire is assembled under the HTS stack (two LTS wire grooves in each superconducting stack groove, and the welding length of HTS stack

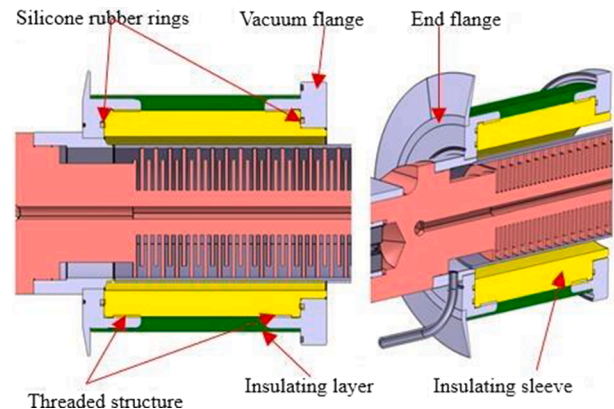


Fig. 4. The structure of insulation flange.

Table 2
Design parameter of HTS section.

Parameter	Value
Operating temperature (K)	5 to 60
Effective length (mm)	413
HTS material	Bi-2223 AgAu
Quantity of 5 fold stacks	36
Critical current (single tape) at self, 77 K (A)	>170
Tape width/thickness (mm)	4.3/0.24

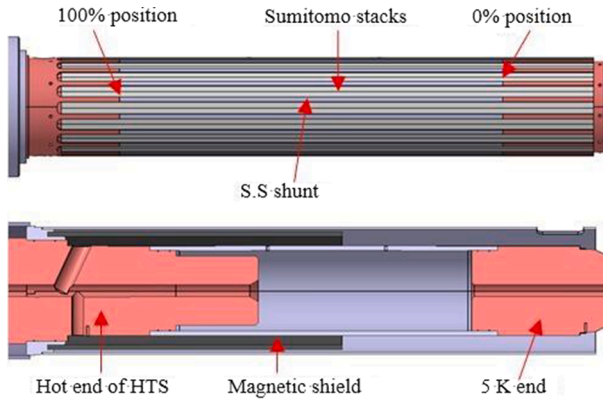


Fig. 5. HTS module of 13 kA current lead.

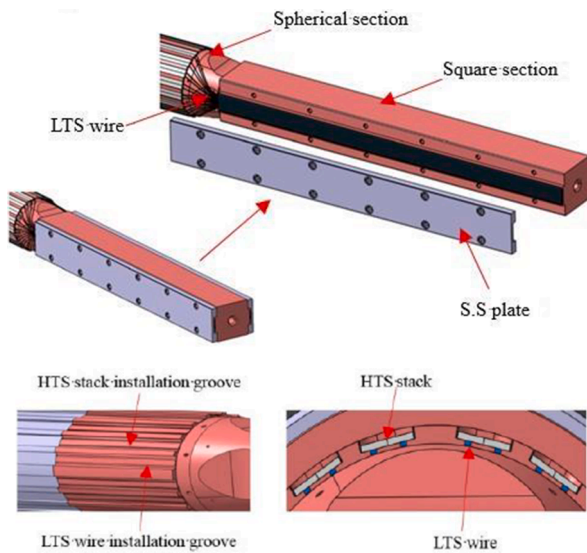


Fig. 6. LTS module of 13 kA current lead.

Table 3
Parameter of LTS wire.

Parameter	Value
Critical current at 4.2 K (A)	1000
Diameter of LTS wire (mm)	0.8
LTS wire material	NbTi
Ratio of 5 copper to non-copper	3:1

and LTS wire is 70 mm), and the other side is soldered on oxygen-free high-conduction copper and tightly fixed with stainless steel cover plate to form LTS joint. This structure integrates the cold end of the HTS module, the LTS module and the LTS joint, which not only reduces the joint resistance, but also simplifies the processing and installation

process of the user, and effectively ensures the operation safety of the HTS current lead.

3. Test setup

A pair of 13 kA current leads was installed into a 0.8 m diameter vertical cryostat which contains an LN2 aluminum shield. The 5 K ends of the two leads were connected by a U-shaped LTS bus-bar with LTS joints. The inlet of 5 K was connected to the cryogenic system, and the 5 K outlet was divided into three channels, two of which flowed to the heat exchanger of the leads and returned to the cryogenic system from the RT terminal, and the other directly returned to the cryogenic system from the RT terminal, and the other directly returned to the cryogenic system (see Fig. 7). The RT terminals of the current leads were connected to a 13.5 kA DC power supply with flexible bus bars. The 900 W/4.5 K cryogenic refrigerator supplied the 5 K helium gas for the two current leads. The leads were cooled and instrumented with temperature sensors, pressure sensors, and flow meters, to allow a proper characterization of their performance during the 5 K helium cooling and the HTS section conduction cooling test.

4. Test results

4.1. Steady-State and LOFA test

In the test, a Proportion Integration Differentiation (PID) control mode was adopted for the temperature on the outlet pipeline of the HEX to realize the control of the HEX flow and the temperature of the HTS section hot end. Firstly, set the potential protection threshold and detect the response of quench detection system and emergency switch. Then the current was raised to the target value and the HEX outlet temperature, pressure, RT terminal temperature, HEX voltage, and HEX helium mass flow rate maintained at steady state for about 50 min.

During the LOFA test, the temperature at the hot end and potential of the HTS section rose slowly, so the potential of the HEX was selected as the quench protection. The LOFA test was performed by stopping the HEX helium supply after establishment of steady-state operation at 13 kA, and stopped with the power off (When the voltage of the heat exchanger exceeds the protection threshold, cut off the power supply

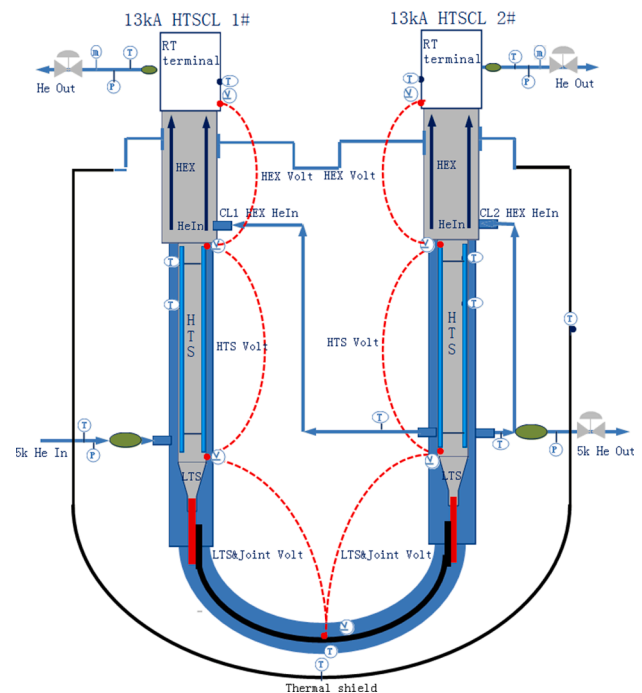


Fig. 7. P&ID of 13 kA current lead.

immediately).

As shown in Table 4, the main parameters were well controlled, in particular the HEX outlet temperature, fixed to 275 K. From Fig. 8, the HEX flows of the current leads under 13 kA steady state test were about 0.63 g/s, and the LOFA time was about 1500 s.

4.2. Joint resistances

To measure the joint resistances a special test was conducted, keeping the 5 K loop and HEX outlet temperature stable and feeding with step current, 2 kA, 4 kA, 6 kA, 8 kA, 10 kA, 13 kA, with a 2 mins flat top for each step. Using special V-taps around the HEX to LTS (R_{CL-HTStop}) and the LTS joint (R_{CL-LTS}). Average the current and voltage reading at give step current. (note that current unit is kilo-Ampere, voltage unit micro-volt). Fit the linear equation for the voltage vs the current then pick up the slop of linear line. The joint resistance will be equal to the slope of fitting curve. Table 5 summarizes the joint resistance results.

4.3. Pulsed current test

The purpose of pulse current test is to further verify the performance of 13 kA current lead. A major goal is to test the ability of automatic flow regulation of lead heat exchanger for a given regulation. Considering that the experimental power supply can't provide a negative current, a dedicated current scenario is used in this test as shown in Fig. 9. It includes fast ramps from flat-top to zero and back at 1 kA/s. In the test, the flow of HTS current lead heat exchanger is successfully controlled at 0.38 g/s under 0 current and 0.57 g/s under 10 kA current under PID automatic regulation.

4.4. Other tests

1) The insulation was tested under 5 kV high voltage. The leak current is <0.2 μ A.

2) In the pressure test of HEX, apply 6.5 bara pressure to the heat exchanger and keep it for 15 min, and the leakage rate is 2.1×10^{-10} Pa·m³/s. See Table 6 for all test results.

5. Conclusion

The low-temperature electrical performance test of 13 kA current lead was completed. The flow requirements, joint resistance, high-voltage insulation and LOFA test results met the design requirements, and the design scheme and manufacturing process were verified. Associated insulation and joint technology were applied in 100 kA current lead.

CRediT authorship contribution statement

Yujun Dong: Conceptualization, Data curation, Methodology, Writing – original draft. **Qingxiang Ran:** Conceptualization, Resources. **Kun Lu:** Methodology, Writing – review & editing. **Jinxing Zheng:** Writing – review & editing. **Chenglian Liu:** Data curation, Investigation, Resources, Writing – review & editing. **Huajun Liu:** . **Kaizhong Ding:**

Table 4
Key parameter status at 13KA.

Parameter	CL1	CL2
Operating Current (kA)	13	13
Voltage of HEX (mV)	35.15	35.94
Operating temperature of top of HTS (K)	<35	<35
Pressure of HEX (Bara)	1.75	1.75
Operating temperature of outlet of HEX (K)	275.05	274.95
Mass flow of HEX under 13 kA (g/s)	0.62	0.63

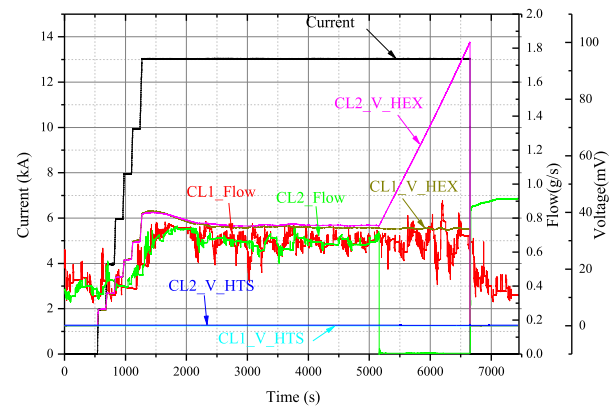


Fig. 8. Key parameter profiles of the Steady-state and LOFA test.

Table 5
Joint resistance summary.

CL	R _{CL-HTStop}	R _{CL-LTS}
CL1	0.2 n Ω	0.8 n Ω
CL2	0.3 n Ω	0.9 n Ω

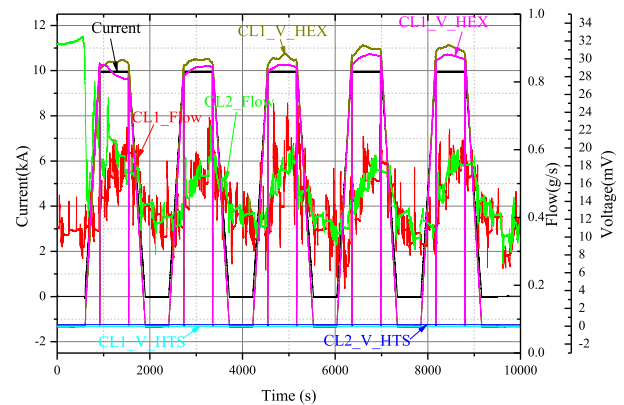


Fig. 9. Flow profile under PID automatic regulation during the pulsed current test.

Table 6
Main performance comparison between experimental results and theoretical design value.

Test items	Test value	Design parameters
Current (kA)	13	13
Operating temperature of top of HTS (K)	<35	<60
Mass flow of HEX under nominal current (g/s)	0.63	0.6–0.8
LOFA time (s)	1500	>120
5 K heat load (W)	0.6	< 3.5
Maximum design voltage (kV)	5	>1.5
Joint resistance (n Ω)	1.2	15
Leakage rate of of HEX (Pa·m ³ /s)	2.1×10^{-10} @6.5 Bara	/

Investigation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

No data was used for the research described in the article.

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